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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the October/November 2008 question paper

0625 PHYSICS

0625/02 Paper 2 (Core Theory), maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2008 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.

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NOTES ABOUT MARK SCHEME SYMBOLS AND OTHER MATTERS

B marks are independent marks, which do not depend on any other marks. For a B mark to be scored, the point to which it refers must actually be seen in the candidate's answer.

M marks are method marks upon which accuracy marks (A marks) later depend. For an M mark to be scored, the point to which it refers **must** be seen in a candidate's answer. If a candidate fails to score a particular M mark, then none of the dependent A marks can be scored.

C marks are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, provided subsequent working gives evidence that they must have known it. e.g. if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows he knew the equation, then the C mark is scored.

A marks are accuracy or answer marks which either depend on an M mark, or which are one of the ways which allow a C mark to be scored.

c.a.o. means "correct answer only".

e.c.f. means "error carried forward". This indicates that if a candidate has made an earlier mistake and has carried his incorrect value forward to subsequent stages of working, he may be given marks indicated by e.c.f. provided his subsequent working is correct, bearing in mind his earlier mistake. This prevents a candidate being penalised more than once for a particular mistake, but **only** applies to marks annotated "e.c.f."

e.e.o.o. means "each error or omission".

brackets ( ) around words or units in the mark scheme are intended to indicate wording used to clarify the mark scheme, but the marks do not depend on seeing the words or units in brackets e.g. 10 (J) means that the mark is scored for 10, regardless of the unit given.

underlining indicates that this must be seen in the answer offered, or something very similar.

OR/or indicates alternative answers, any one of which is satisfactory for scoring the marks.

Spelling Be generous about spelling and use of English. If an answer can be understood to mean what we want, give credit.

Significant figures Answers are acceptable to any number of significant figures $\geqslant 2$, except if specified otherwise, or if only 1 sig. fig. is appropriate.

Units It is expected that all final answers will have correct units. Deduct one unit penalty for each incorrect or missing unit, maximum 1 per question. No unit penalty if unit is missing from final answer but is shown correctly in the working.

Fractions These are only acceptable where specified.

Extras Ignore extras in answers if they are irrelevant; if they contradict an otherwise correct response or are forbidden by mark scheme, use right + wrong = 0

Ignore Indicates that something which is not correct is disregarded and does not cause a right plus wrong penalty.

Not/NOT Indicates that an incorrect answer is not to be disregarded, but cancels another otherwise correct alternative offered by the candidate i.e. right plus wrong penalty applies.

Work which has been crossed out, but not replaced, should be marked as if it had not been crossed out.

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1	**(a)**	attempt at subtraction		C1	
		3 hrs 20 mins		C1	
		200 mins		A1	
	(b)	200/20 OR 200/10	e.c.f. from **(a)**	C1	
		10 OR 20	e.c.f.	A1	[5]
2	**(a)**	$L \times B \times H$		C1	
		$2 \times 1.5 \times 1$		C1	
		3 (m^3)		A1	
	(b)	$M = V \times D$		C1	
		3000 (kg)	e.c.f.	A1	
		correct units in both **(a)** and **(b)**		B1	[6]
3	**(a)**	YZ OR WZ and WY OR XZ and XY		B1	
	(b)	**(i)**	4.2 – 4.6 (N)	B1	
		(ii)	his **(i)**	B1	
			N	B1	
		(iii)	increases	B1	
		(iv)	goes up OR oscillates OR accelerates	B1	[6]
4	**(a)**	radio OR TV		B1	
		ultra-violet		B1	
	(b)	long wavelength written at LH end		B1	
	(c)	any three acceptable uses		B1,B1,B1	
	(d)	red		B1	
		violet (NOT blue, purple, mauve)		B1	[8]

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5 **(a)** **(i)** reflected B1

(ii) same frequency B1

(b) **(i)** 480 (m) B1

(ii) distance/time C1
his**(i)**/1.6 C1
300 (m/s) e.c.f. A1

(iii) measured distance incorrect)
measured time incorrect OR effect of reaction time) any 2 B1,B1
any reference to temperature)
any reference to wind)
any reference to humidity)
any reference to pressure)
time interval very small to measure on a stopwatch)

[8]

6 **(a)** distance approximately marked (±5 mm) C1
distance accurately marked (±2 mm) A1

(b) **(i)** falls B1

(ii) air mols bombard surfaces B1
cause pressure on Hg B1

(c) Hg at same height as in LH tube (by eye) B1

(d) rises rises) B1 × 4
falls stays the same)

[10]

7 **(a)** **(i)** normal correct, by eye B1

(ii) reflected ray in correct direction, by eye (condone poor normal) B1

(iii) *i* correctly labelled (condone poor normal) B1

(iv) *r* correctly labelled (condone poor normal) B1

(b) $i = r$ B1

(c) **(i)** second mirror correct, by eye B1

(ii) ray parallel to original ray B1 [7]

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8	**(a)**	**(i)**	N at left and S at right		B1	
		(ii)	attract e.c.f.		B1	
	(b)	**(i)**	N at left and S at right		B1	
		(ii)	attract e.c.f.		B1	
	(c)		attract		B1	
	(d)		nothing		B1	[6]
9	**(a)**		a		B1	
	(b)		B		B1	
	(c)		I = V/R		C1	
			6/4		A1	
			1.5		A1	
			A OR amp(s) OR ampere(s)		B1	
	(d)		connection 2		B1	
			both receive full voltage OR both run at full brightness		B1	
			if one blows/switched off, other will continue working		B1	[9]
10	**(a)**		contact position shown at centre of potential divider, by eye		B1	
	(b)		current in coil core magnetised armature attracted armature pivots armature pushes contacts closed	) any 3	B1,B1,B1	[4]

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11	**(a)**	core		B1	
	(b)	a.c.		B1	
	(c)	$V_P/V_S = N_P/N_S$ in any form		C1	
		correct substitution		C1	
		200		A1	
	(d)	voltage too high OR lamp will blow		B1	[6]
12	**(a)**	84		B1	
	(b)	218		B1	
	(c)	**(i)**	2	B1	
		(ii)	4 allow both marks for interchanged answers for **(i)** and **(ii)**, if answers to **(a)** and **(b)** were interchanged	B1	
		(iii)	alpha-particle	B1	[5]